



SQL Project on

PIZZA SALES

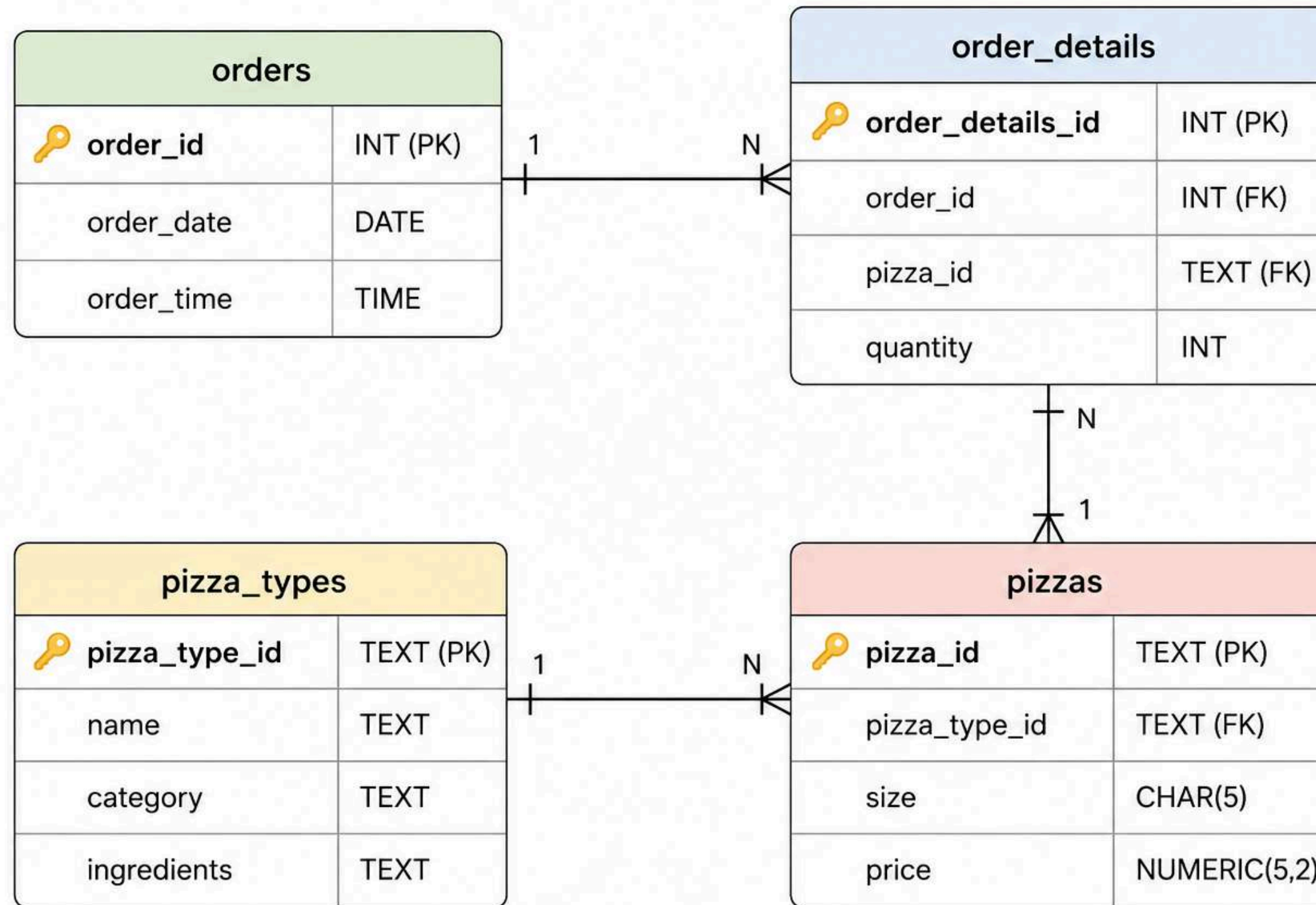
Note: This is a learning project and dataset is not real.

ABOUT THIS PROJECT

Hello, My name is Priyanshu Patel. This project analyzes a pizza sales dataset using SQL to extract meaningful business insights. The dataset consists of four related tables: Orders, Order Details, Pizzas, and Pizza Types. The analysis focuses on answering business questions related to sales performance, customer ordering behavior, revenue generation, and product popularity by using SQL concepts such as Joins, Aggregate Functions, GROUP BY, ORDER BY, Subqueries, and Window Functions.



PIZZA SALES - DATA MODEL



1. RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT  
    COUNT(*) AS total_orders  
FROM orders;
```

	total_orders 
1	21350



2. CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT  
    SUM(od.quantity * p.price) AS total_revenue  
FROM order_details od  
INNER JOIN pizzas p  
ON od.pizza_id = p.pizza_id;
```

	total_revenue numeric
1	817860.05



3. IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    pt.name,
    p.price
FROM pizza_types pt
JOIN pizzas p
ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

	name text	price numeric (5,2)
1	The Greek Pizza	35.95



4. IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
  p.size AS size_of_pizza,
  COUNT(od.order_details_id) AS order_count
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY COUNT(od.order_details_id) DESC
LIMIT 1;
```

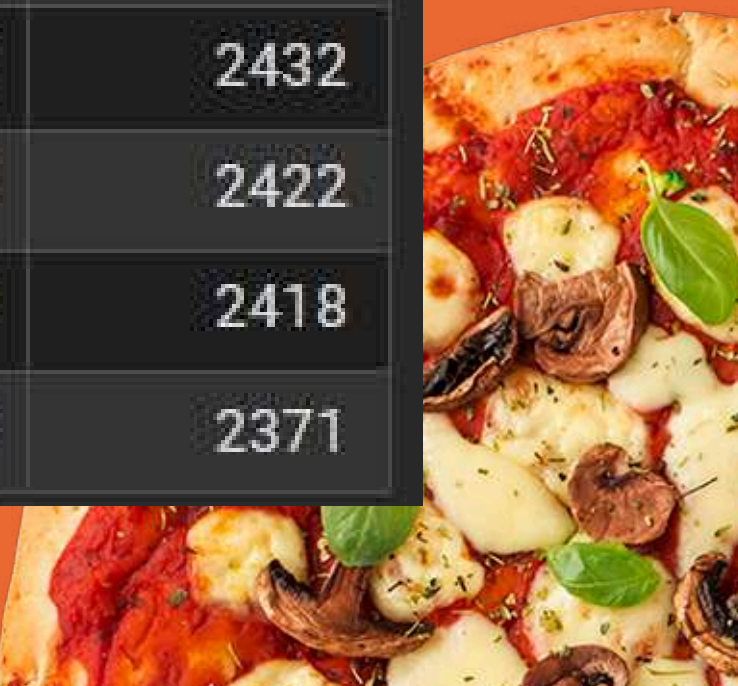
	size_of_pizza character (5) 🔒	order_count bigint 🔒
1	L	18526



5. LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pt.name,
    COUNT(od.order_details_id) AS orders,
    SUM(od.quantity) as quantity
FROM pizzas p
JOIN order_details od
ON p.pizza_id = od.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY quantity DESC
LIMIT 5;
```

	name 	orders 	quantity 
	text	bigint	bigint
1	The Classic Deluxe Pizza	2416	2453
2	The Barbecue Chicken Pizza	2372	2432
3	The Hawaiian Pizza	2370	2422
4	The Pepperoni Pizza	2369	2418
5	The Thai Chicken Pizza	2315	2371



6. JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT
    DISTINCT pt.category,
    SUM(od.quantity) AS total_quantity
FROM pizzas p
join order_details od
ON p.pizza_id = od.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP by pt.category
ORDER BY total_quantity DESC;
```

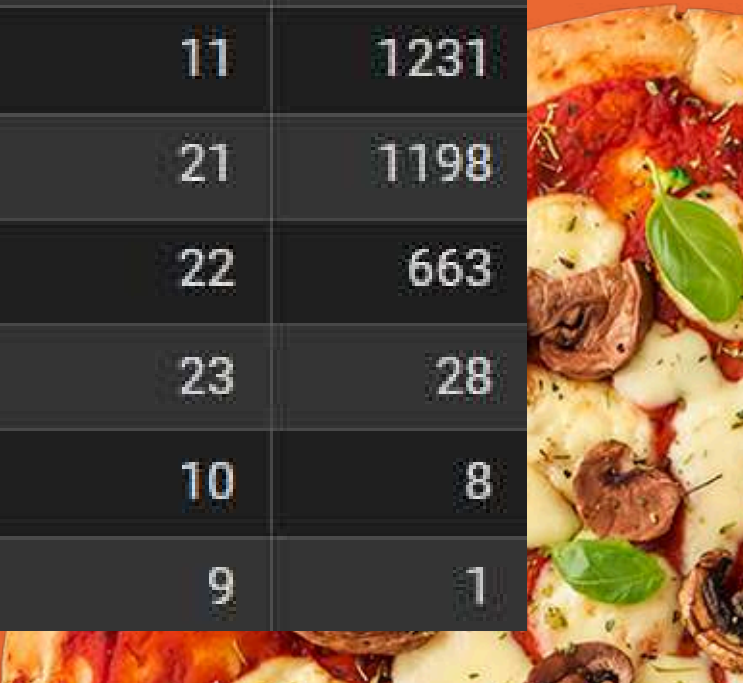
	category 	total_quantity 
	text	bigint
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050



7. DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    DATE_PART('HOUR', order_time) AS hours,
    COUNT(order_id) AS orders
FROM orders
GROUP BY hours
ORDER BY orders DESC;
```

	hours double precision 	orders bigint 
1	12	2520
2	13	2455
3	18	2399
4	17	2336
5	19	2009
6	16	1920
7	20	1642
8	14	1472
9	15	1468
10	11	1231
11	21	1198
12	22	663
13	23	28
14	10	8
15	9	1



8. JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT
    category,
    COUNT(name) pizza_count
FROM pizza_types
GROUP BY category;
```

	category text 🔒	pizza_count bigint 🔒
1	Supreme	9
2	Chicken	6
3	Classic	8
4	Veggie	9



9. GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```
SELECT TRUNC(AVG(total_order_qty), 0) AS avg_pizza_ordered_per_day
FROM
  (SELECT
    o.order_date AS "Date",
    SUM(od.quantity) AS total_order_qty
  FROM orders o
  JOIN order_details od
  ON o.order_id = od.order_id
  GROUP BY "Date") AS t;
```

	avg_pizza_ordered_per_day 
1	138



10. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT pt.name AS pizza_type, SUM(od.quantity * p.price) AS revenue
FROM pizzas p
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
JOIN order_details od
ON p.pizza_id = od.pizza_id
GROUP BY pizza_type
ORDER BY revenue DESC
LIMIT 3;
```

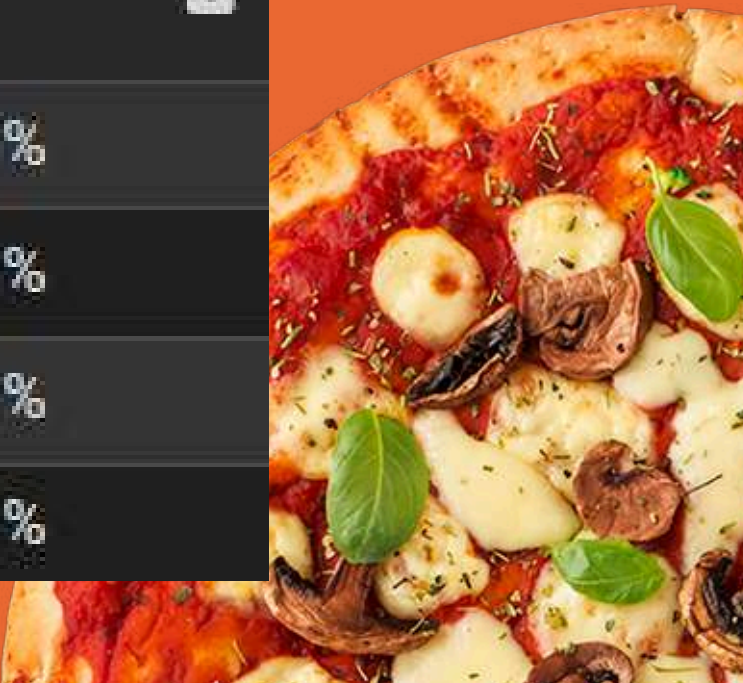
	pizza_type text	revenue numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768.00
3	The California Chicken Pizza	41409.50



11. CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT
  pt.category AS pizza_category,
  TRUNC((SUM(od.quantity * p.price) /
    (SELECT
      SUM(od.quantity * p.price) AS total_revenue
    FROM order_details od
    INNER JOIN pizzas p
    ON od.pizza_id = p.pizza_id)) * 100, 2) || ' %' AS revenue_pct
FROM pizzas p
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
JOIN order_details od
ON p.pizza_id = od.pizza_id
GROUP BY pizza_category
ORDER BY revenue_pct DESC;
```

	pizza_category 	revenue_pct 
	text	text
1	Classic	26.90 %
2	Supreme	25.45 %
3	Chicken	23.95 %
4	Veggie	23.68 %



12. ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT order_date,  
SUM(revenue) OVER(ORDER BY order_date) AS cum_revenue  
FROM  
  (SELECT  
    o.order_date AS order_date,  
    SUM(od.quantity * p.price) AS revenue  
  FROM pizzas p  
  JOIN order_details od  
  ON p.pizza_id = od.pizza_id  
  JOIN orders o  
  ON od.order_id = o.order_id  
  GROUP BY o.order_date) AS sales;
```

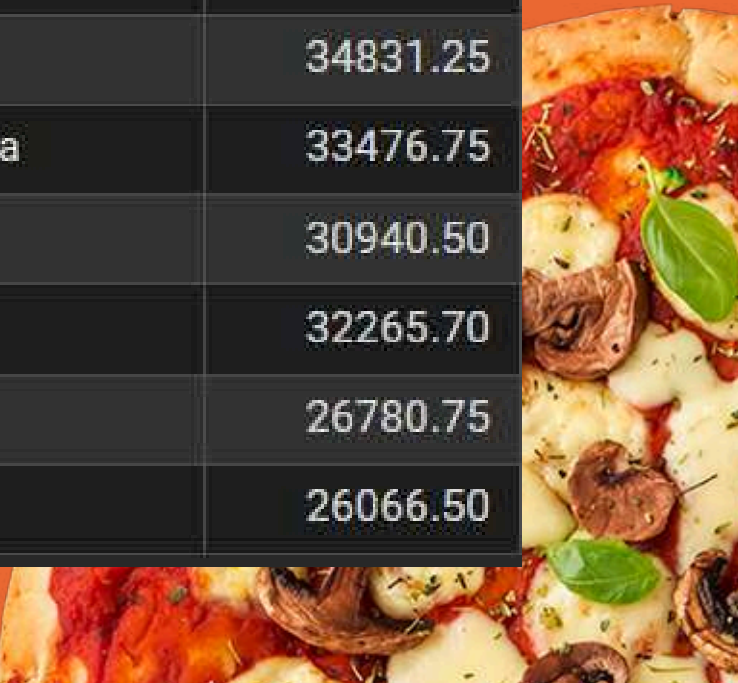
	order_date 	cum_revenue  numeric
1	2015-01-01	2713.85
2	2015-01-02	5445.75
3	2015-01-03	8108.15
4	2015-01-04	9863.60
5	2015-01-05	11929.55
6	2015-01-06	... 14358.50
7	2015-01-07	16560.70
8	2015-01-08	19399.05



13. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
SELECT category, name, revenue
FROM
  (SELECT
    category, name, revenue,
    RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn
  FROM
    (SELECT
      pt.category,
      pt.name,
      SUM(od.quantity * p.price) AS revenue
    FROM pizzas p
    JOIN order_details od
    ON p.pizza_id = od.pizza_id
    JOIN pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
    GROUP BY pt.category, pt.name
    ) AS t1
  ) AS t2
WHERE rn <= 3;
```

	category 	name 	revenue 
	text	text	numeric
1	Chicken	The Thai Chicken Pizza	43434.25
2	Chicken	The Barbecue Chicken Pizza	42768.00
3	Chicken	The California Chicken Pizza	41409.50
4	Classic	The Classic Deluxe Pizza	38180.50
5	Classic	The Hawaiian Pizza	32273.25
6	Classic	The Pepperoni Pizza	30161.75
7	Supreme	The Spicy Italian Pizza	34831.25
8	Supreme	The Italian Supreme Pizza	33476.75
9	Supreme	The Sicilian Pizza	30940.50
10	Veggie	The Four Cheese Pizza	32265.70
11	Veggie	The Mexicana Pizza	26780.75
12	Veggie	The Five Cheese Pizza	26066.50



THANK
YOU

